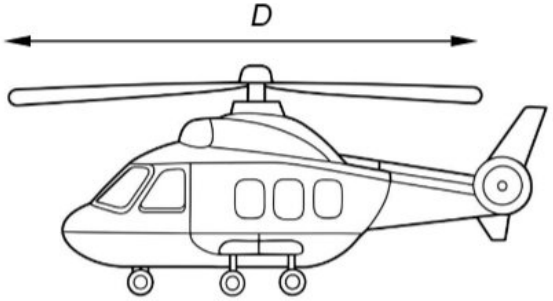
6 A helicopter hovers stationary in mid-air. Its rotor blades sweep out a circular path of diameter D and push down a continuous column of air of density ρ at a uniform speed v. The air above the blades is taken to be stationary.

Which expression correctly represents the average power P delivered to the air column by the blades?



A $\frac{1}{8}\pi\rho D^2 v^3$ **B** $\frac{1}{2}\pi\rho D^2 v^3$ **C** $\frac{1}{4}\pi\rho D^2 v^3$ **D** $\frac{1}{4}\pi\rho D^2 v^2$